

Relative Probability Matrices for Predicting Outcomes in the Indian Premier League

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Abstract

Predicting match outcomes in the Indian Premier League (IPL) is a challenging problem due to variations in team performance across seasons. In this work, we present a Relative Probability Matrix (RPM) approach that uses historical match outcomes to estimate win probabilities between any two teams. Our method builds season-specific RPMs and averages them to form a long-term probability table. We then combine the long-term and seasonal RPMs to predict match outcomes. Evaluated on IPL datasets from 2008–2024, our framework provides a transparent and reproducible method for cricket analytics.

1 Introduction

Traditional methods for sports outcome prediction, such as Elo ratings or logistic regression, often fail to capture direct head-to-head dynamics between teams. We instead use Relative Probability Matrices (RPMs), which explicitly encode pairwise winning percentages between all IPL teams. This method allows simple, interpretable, and data-driven predictions without relying on black-box machine learning models.

2 Methodology

2.1 Data Collection

We use IPL match results from 2008–2024. Preprocessing steps include team name standardization and removal of incomplete matches.

2.2 Relative Probability Matrix (RPM)

For each season s , we construct a square matrix $M^{(s)}$, where each entry represents the empirical win percentage of team i over team j :

$$M_{ij}^{(s)} = \frac{\text{wins of team } i \text{ against } j}{\text{total matches between } i \text{ and } j} \times 100$$

Diagonal entries are set to 100, since a team always has 100% against itself. If two teams have never played, their entries are initialized as 0.

2.3 Long-Term Aggregation

We maintain an overall probability matrix by averaging across all seasonal RPMs:

$$M^{(L)} = \frac{1}{S} \sum_{s=1}^S M^{(s)}$$

where S is the number of seasons considered.

2.4 Hybrid Prediction

To predict the outcome of a match between team A and team B , we combine:

- Current season RPM ($M^{(S)}$)
- Long-term RPM ($M^{(L)}$)

The combined probability is the simple average:

$$\hat{p}_{AB} = \frac{M_{AB}^{(S)} + M_{AB}^{(L)}}{2}$$

If $\hat{p}_{AB} > 50\%$, we predict team A as the winner; otherwise team B .

2.5 Implementation

The method is implemented in Python using Pandas and NumPy. Predictions are stored in `Probs.csv` for further evaluation.

3 Results

3.1 Example RPM

A sample probability matrix is shown below:

	CSK	MI	RCB
CSK	100.0	57.1	71.4
MI	42.9	100.0	50.0
RCB	28.6	50.0	100.0

This indicates, for example, that CSK has historically won 57.1% of matches against MI and 71.4% against RCB.

3.2 Hybrid Predictions

The algorithm produces win probabilities for each season. For example:

- 2020: Predicted CSK > MI with 54% probability.
- 2021: Predicted RCB > KKR with 52% probability.

(All predictions are stored for evaluation.)

4 Discussion

Our RPM-based approach offers:

- **Transparency:** Directly interpretable win probabilities.
- **Simplicity:** No machine learning models are required.
- **Flexibility:** Works across all IPL seasons.

Limitations:

- Does not handle contextual factors such as toss or venue.
- Probabilities are based only on historical win ratios, without smoothing.
- If teams have never met before, probabilities default to 0.

5 Conclusion

We proposed a simple Relative Probability Matrix framework for predicting IPL outcomes. By combining seasonal and long-term win percentages, the method produces reasonable probabilistic forecasts. Future extensions may include smoothing, adaptive weighting, or contextual corrections to improve accuracy.